

Yuyao Yang

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RESEARCH INTERESTS

Areas: Machine Learning; Large Language Models; Vision-Language Models; Computer Vision; Latent Reasoning; Agentic AI

EDUCATION

University of Illinois Chicago <i>Ph.D. in Computer Science Advisor: Prof. Philip S. Yu</i>	Aug 2024-Present
Cranfield University <i>M.Sc. in Applied Artificial Intelligence Thesis: Data Fusion Versus Data Integration</i>	Sep 2022-Sep 2023
Zhengzhou University <i>B.Eng. in Software Engineering</i>	Sep 2015-Jul 2019

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Illinois Chicago <i>Big Data and Social Computing Lab Advisor: Prof. Philip S. Yu</i>	Aug-Dec 2024
Conducted collaborative research across large language models, agentic RAG, vision-language models, and multimodal machine learning.	

TEACHING EXPERIENCE

Teaching Assistant, University of Illinois Chicago <i>Department of Computer Science</i>	2024-2026
- CS 141 - Program Design II: C++ programming, program design, and implementation. - CS 151 - Mathematical Foundations of Computing: discrete mathematical concepts for computing. - CS 412 - Introduction to Machine Learning: classification, density estimation, clustering, and online learning.	

SELECTED RESEARCH PROJECTS

Every Head Counts: Mask-Alignment Calibration for Semantic Recovery Under Domain Shift <i>Ongoing Research</i>	
- Developing a training-free calibration framework that assigns continuous reliability weights to VLM attention heads, converting noisy class-token attention into reusable class-specific spatial priors without updating the VLM or training a decoder. - Building and evaluating a semantic-recovery pipeline that combines calibrated softmaps with frozen-segmenter features to restore classes missed under night, fog, rain, and other visual domain shifts. - Evaluating attention localization and downstream recovery across clean and adverse-condition benchmarks using frozen VLMs and multiple segmentation architectures.	
Latent Reasoning with Embedded Language Flows (ELF) <i>Ongoing Research</i>	
- Investigating flow-based latent reasoning that represents intermediate computation as continuous trajectories and iteratively refines multiple latent-thought slots. - Synthesizing work on continuous thought, flow-based reasoning, adaptive computation, and latent world models to identify practical directions for whole-trajectory refinement. - Designing a minimum validation protocol with no-chain-of-thought, text-chain-of-thought, and deterministic latent-reasoning baselines, plus accuracy-compute and causal-intervention tests.	

ACADEMIC SERVICE

Peer Reviewer: ACL; EMNLP

TECHNICAL SKILLS

Programming: Python; C++
Frameworks and Models: PyTorch; Transformers; Large Language Models; Vision-Language Models
Methods: Machine Learning; Computer Vision; Multimodal Learning; Agentic and Latent Reasoning

SELECTED PUBLICATIONS

1. Yangning Li, Weizhi Zhang, **Yuyao Yang**, Wei-Chieh Huang, Yaozu Wu, Junyu Luo, Yuanchen Bei, Henry Peng Zou, Xiao Luo, Yusheng Zhao, Chunkit Chan, Yankai Chen, Zhongfen Deng, Yinghui Li, Hai-Tao Zheng, Dongyuan Li, Renhe Jiang, Ming Zhang, Yangqiu Song, and Philip S. Yu. **"A Survey of RAG-Reasoning Systems in Large Language Models."** *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 12120-12145.
2. Chunyu Miao, Henry Peng Zou, Yangning Li, Yankai Chen, Yibo Wang, Fangxin Wang, Yifan Li, Wooseong Yang, Bowei He, Xinni Zhang, Dianzhi Yu, Hanchen Yang, Hoang H. Nguyen, Yue Zhou, Jie Yang, Jizhou Guo, Wenzhe Fan, Chin-Yuan Yeh, Panpan Meng, Liancheng Fang, Jinhu Qi, Wei-Chieh Huang, Zhengyao Gu, Yuwei Han, Langzhou He, **Yuyao Yang**, Yinghui Li, Hai-Tao Zheng, Xue Liu, Irwin King, and Philip S. Yu. **"RECODE-H: A Benchmark for Research Code Development with Interactive Human Feedback."** *International Conference on Learning Representations (ICLR)*, 2026.
3. Zhongfen Deng, Yibo Wang, Yueqing Liang, Jiangshu Du, **Yuyao Yang**, Liancheng Fang, Langzhou He, Yuwei Han, Yuanjie Zhu, Chunyu Miao, Weizhi Zhang, Jiahua Chen, Yinghui Li, Wenting Zhao, and Philip S. Yu. **"A Survey of Multimodal Models on Language and Vision: A Unified Modeling Perspective."** *Data Mining and Machine Learning*, 1(1), Article 100001, 2025. Equal contribution.
4. Weizhi Zhang, Yangning Li, Yuanchen Bei, Junyu Luo, Guancheng Wan, Liangwei Yang, Chenxuan Xie, **Yuyao Yang**, Wei-Chieh Huang, Chunyu Miao, Henry Peng Zou, Xiao Luo, Yusheng Zhao, Yankai Chen, Chunkit Chan, Peilin Zhou, Xinyang Zhang, Chenwei Zhang, Jingbo Shang, Ming Zhang, Yangqiu Song, Irwin King, and Philip S. Yu. **"From Web Search towards Agentic Deep Research: Incentivizing Search with Reasoning Agents."** *arXiv:2506.18959*, 2025. Preprint.